Self-assessment Worksheet (Worksheet 3)

Do I understand this weeks material?

Real Analysis Topics	Tick the box when you feel confident with each topic
• Understand the proofs of the algebraic limit theorem.	☐
• Use the algebraic limit theorem to compute limits of specific sequences.	☐
• Use and understand the squeeze theorem.	☐
• Understand and know the definition of a monotone sequence and subsequence.	☐
• Utilise the monotone convergence theorem to prove the convergence of specific sequences.	☐
• Know and understand the Bolzano-Weierstrauss theorem.	☐

Exercises and Questions

Unless stated otherwise, you may assume any results proven in lecture.

Exercises and questions are organised and classified into three groups:

- **Starter Questions** - these questions will help cement the basic techniques.
- **Challenge Questions** - these questions are more challenging, designed to test your deeper understanding of the topic.
- **Extension Questions** - questions to try yourself in your own time to extend your knowledge and understanding.

Starter Qs

Have a go at these "warm-up" questions to get you started.

Question 1 I can do this question ☐

- Using the algebraic limit theorem, write down the limits of (a_n) give that:

  (a) $a_n = \dfrac{1}{n} + \dfrac{6}{n^3}$.

  (b) $a_n = \dfrac{n^2 - 5}{n^2 + 6n}$.

  (c) $a_n = \dfrac{5n^2 + 3n - 2}{2n^2 + 7n + 3}$.

  (d) $a_n = \dfrac{-2n^3 + 4n^2 + 1}{7n^3 + 6n^2 + 5n + 4}$.

  (e) $a_n = \dfrac{n^5 + n^3 + n}{n^7 + n^5 + n^3 + n}$.

  Make sure to state which part of the algebraic limit theorem you used in each step.

Question 2 I can do this question ☐

- Give an example of a convergent sequence that is not monotone. Additionally, give an example of a monotone sequence that is not convergent.

Question 3 I can do this question ☐

- When does a monotone increasing sequence converge? When does a monotone decreasing sequence converge?

Question 4 I can do this question ☐

- Let (a_n) and (b_n) be divergent sequences. Give example of sequences (a_n) and (b_n) such that the following limits exist. If no such example exists, say the limit cannot exist.
  (a) $\lim(ca_n)$ for all $c \in \mathbb{R}$.
  (b) $\lim(a_n + b_n)$.
  (c) $\lim(a_n b_n)$
  (d) $\lim\left(\frac{a_n}{b_n}\right)$ with $b_n \neq 0$ for all $n \in \mathbb{N}$.

Challenge Qs

Now you are feeling more confident try these more challenging questions.

Question 5 I can do this question ☐

- Let (a_n) and (b_n) be sequences that both diverge to positive infinity. Prove or disprove the following:
  (a) $\lim(ca_n) = \begin{cases} \infty, \text{ if } c \in \mathbb{R}^+ \\ -\infty, \text{ if } c \in \mathbb{R}^- \end{cases}$.
  (b) $\lim(a_n + b_n) = +\infty$.
  (c) $\lim(a_n b_n) = +\infty$.
  (d) $\lim\left(\frac{a_n}{b_n}\right) = k \in \mathbb{R}$, providing $b_n \neq 0$ for all $n \in \mathbb{N}$.

Question 6 I can do this question ☐

- Using the squeeze theorem, prove the following limits exist and find them explicitly.
  (a) $\lim\left(\frac{\sin(2n)}{n}\right)$.
  (b) $\lim\left(\frac{\sin(2n)}{n^2\cos(n)}\right)$.
  (c) $\lim\left(\frac{\ln(n)}{n}\right)$.

Question 7 I can do this question ☐

- Using Corollary 8.4, prove that (a_n) is divergent with
  (a) $a_n = 1 + (-1)^n$.
  (b) $a_n = \cos(n\pi)$.

Question 8 I can do this question ☐

- Prove Theorem 6.3 part (ii) and (iii) from the notes.

Question 9 I can do this question ☐

- Prove the if ($\Leftarrow$) statement in Theorem 8.3 in the notes.

Question 10 I can do this question ☐

- Let (b_n) be a monotone increasing sequence with $b_1 \geq 0$. Define a new sequence (a_n) by

$$a_n = \frac{2b_n + 3}{b_n + 2}.$$

  (a) By considering $a_{n+1} - a_n$, show that (a_n) is monotone increasing.

  (b) By considering $2 - a_n$, prove that (a_n) is bounded and hence show it is convergent. Write down the value it converges to.

Question 11 I can do this question ☐

- State the Bolzano-Weierstrass theorem.

  (a) Give an example of a sequence (a_n), where $6 \leq a_n \leq 8$ for every $n \in \mathbb{N}$ and (a_n) is divergent. Using this example, write down a subsequence of (a_n) that converges.

  (b) Let (b_n) be the sequence given by

$$b_n = \frac{3n\cos(2n^2 + 1)}{n + 2}.$$

  Explain why (b_n) has a convergent subsequence.

Question 12 I can do this question ☐

- Given an example of a sequence (a_n) with:

  (a) A subsequence that converges to 1,

  (b) Another subsequence that converges to 17,

  (c) And a final subsequence that converges to π.

Question 13 I can do this question ☐

- (a) Prove that the sequence

$$(a_n) = (\sqrt{2}, \sqrt{2 + \sqrt{2}}, \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \dots)$$

  converges and find its limit.

  (b) Does the sequence

$$(b_n) = (\sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \dots)$$

  converge? If so, find its limit.

Question 14 I can do this question ☐

- Prove that the sequence (a_n) with $a_n = x^n$ converges to 0 if $x \in (-1, 1)$, converges to 1 if $x = 1$, and diverges otherwise.

Question 15 I can do this question ☐

- Let (a_n) and (b_n) be Cauchy sequences and let $c_n = |a_n - b_n|$. Prove that (c_n) is also a Cauchy sequence.

Extension Qs

These questions are optional extension problems so you can gain more practice and confidence with these topics.

Question 16 I can do this question ☐

- (a) Using the monotone convergence theorem, prove without the use of the completeness axiom the Archimedean property of the real numbers.
  (b) Using the monotone convergence theorem, prove without the use of the completeness axiom the nested interval property.

Question 17 I can do this question ☐

- Prove that every non-empty subset of the real numbers bounded above has a supremum directly from the Bolzano-Weierstrass theorem.

End Material

Review and Summary

Make sure you have fully understood and read the notes. Once the solutions for this worksheet have been released, check your answers and make a note of areas of concern. If you are still unsure then email your lecturer or drop by in office hours and we will be happy to help!

Completing Worksheets: Tips

- Proceed carefully and systematically from beginning to end making sure to note theorems and properties utilised.
- Work together with other students, discuss ideas and concepts.
- Prioritise carefulness over speed.
- Include enough information in your answers that you can easily understand your thinking when you come back to review the worksheets at a later date.
- Do ask for help if you get stuck!